Data Collection Process

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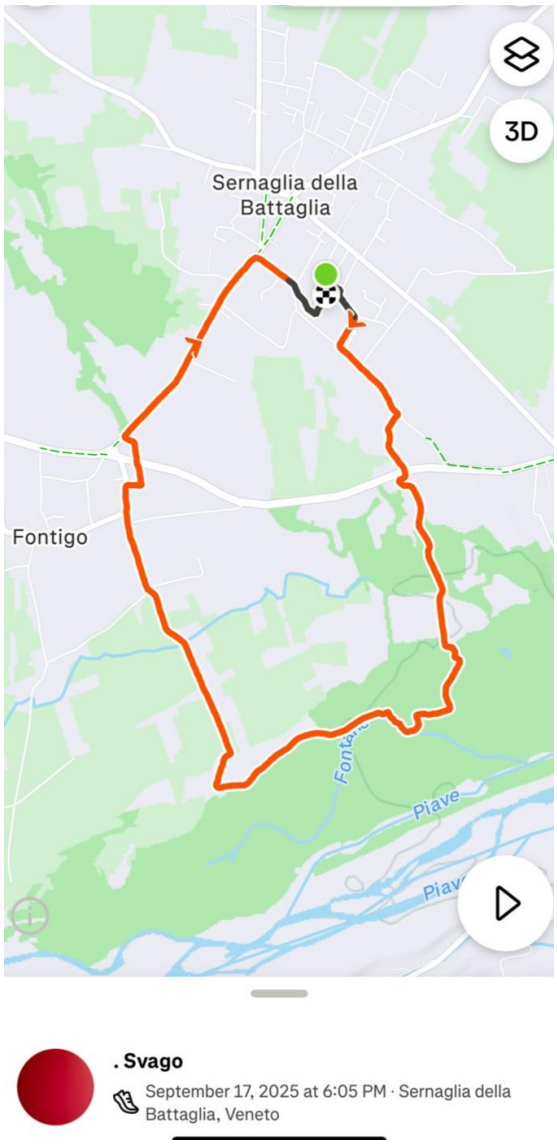
Data Extraction and Ingestion

The raw data for this study was sourced from Andrea's personal Apple Watch, exported via the iOS Health App in a high-volume XML format. Due to the massive scale of the raw telemetry, we developed a specialized Python-based processing engine designed for high-throughput ingestion. By implementing a streaming parsing technique, we processed millions of data points sequentially, ensuring system stability and memory efficiency.

After rigorous quality filtering, the final dataset consists of 43 valid days, spanning approximately five months from August 14, 2025, to January 16, 2026. In our framework, the heart rate profile recorded for each day is treated as a distinct discrete realization of an underlying continuous process. To transition from these discrete observations to the continuous curves required for Functional Data Analysis (FDA), we implemented a smoothing process within the R environment. This step was essential to filter out measurement noise and transform the raw heart rate sequences into actual mathematical functions, providing a robust basis for the subsequent statistical modeling.



Example of physical activity performed by Andrea and recorded via Strava.

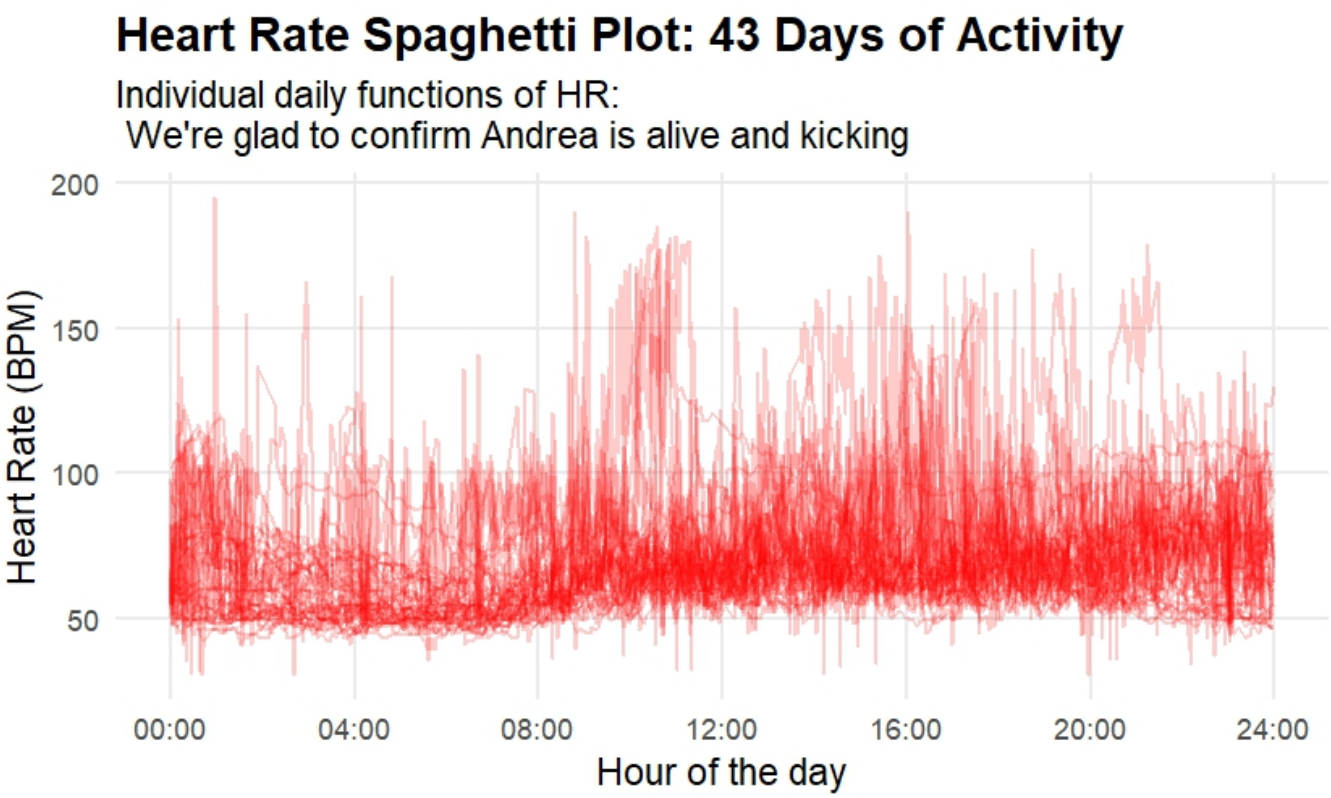
Andrea being Andrea.

Feature Engineering and Encoding

To ensure the dataset was ready for statistical modeling in R, we applied specific encoding logic to the raw categorical strings:

- **Motion Context**: Activity states were mapped to a binary scale (0 = Sedentary, 1 = Active) to clearly distinguish rest periods from physical movement.
- **Sleep Metrics**: We extracted two layers of detail: *SleepStatus* (Binary: Awake/Asleep) and *SleepStage*. The latter was compressed into an ordinal scale (1 = Core, 2 = Deep, 3 = REM) to reflect the physiological intensity and depth of the sleep cycles.
- **Temporal Features**: Every observation was timestamped and decomposed into Year, Month, Day, Hour, and Minute.

Quality Assurance and Physiological Validation

A rigorous validation protocol was established to ensure the integrity of the analysis:

- **Daily Completeness Threshold**:We only retained days with a minimum of 20 hours of recorded data. This ensures that the daily functional profiles are representative and not skewed by long periods of missing observations.
- **Biological Guardrails**:We implemented strict biological constraints. Any heart rate observation outside the realistic human range of 30 to 220 Beats Per Minute (BPM) was discarded as a sensor artifact.